

Shaurya Verma

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Builder of high-agency AI systems for healthcare, accessibility, and automated job discovery.

Final-year CSE student who finds the real problem and directs AI to ship the thing that solves it — 3 shipped systems built end to end and live; a national NTSE scholarship for aptitude; 1st place, Agentic & GenAI Showcase, Techgyan Hackathon, IIT Ropar; built for a place that values logical reasoning.

PROJECTS

Braillix — a refreshable Braille display for mathematics (software half) Aug 2026

The teacher's blackboard on a blind child's fingers: a maths teacher writes, types or photographs a lesson and it lands on refreshable braille cells, live and verified, in English and Hindi — no account, no server, no API key, offline after first load. · [braillix.vercel.app](#)

- Built the on-device recognition pipeline that turns handwriting, typed maths and textbook photos into Nemeth braille — Tesseract for English and Hindi words, a **76 MB** on-device formula model for maths — with nothing leaving the room
- Verified the syllabus live: all **232** curriculum lines from classes **1–12**, arithmetic to Bayes' theorem, translate and read back cleanly in about a second on the production site
- Engineered **3** independent proofs per expression so every dot can be trusted — Nemeth from **speech-rule-engine** over MathML, golden vectors against published tables, and a hand-rolled readback engine that reconstructs meaning from the dots alone
- Reduced press-to-result from **64 s** to about **1.25 s** by warming both recognisers in the background at app start

Sehat Saarthi — Punjabi-first AI screening co-pilot for government hospitals Jun 2026

A Punjabi-first, fully free, multi-disease screening co-pilot for Punjab's government hospitals — the doctor drives, the saarthi guides.

Proof of concept, pending clinical validation. · [sehat-saarthi-punjab.vercel.app](#)

- Trained **9** disease-screening modules on public datasets and deployed only the **7** a doctor can trust after a written reliability analysis — heart disease ROC-AUC **0.948**, chest X-ray **0.957**, breast cancer **0.998**; retinopathy at **0.686** kept out
- Designed the whole product Punjabi-first across **7** live modules: Gurmukhi primary on every element, Web **Speech** read-outs for low-literacy patients, and a printable bilingual referral slip for the paper workflow rural clinics actually use
- Built the three-layer free stack — a Vite frontend on Vercel, FastAPI on a **Hugging Face** Space, tiny .joblib and .onnx models bundled in the package — with a module router so adding a disease is configuration, never a rewrite
- Tuned every model high-recall with a mandatory bilingual disclaimer and an honest recall note; the service degrades to a deterministic stub instead of crashing when a model is missing

Sifarish — a job-hunt chief of staff that refuses to lie | TypeScript, Vite, Dexie, pdf-lib Sep 2026

A personal team of agents for one candidate: an evidence ledger compiled into a truth-checked résumé and cover letter per job posting, lawful worldwide job discovery, an apply cockpit and a pipeline board — fully functional with zero API keys — this résumé was compiled by it · [sifarish-shv-s-projects.vercel.app](#)

- Architected the truth-compiler: every résumé line carries ledger evidence ids, the **LLM** may rephrase but never mint, and the PDF's text layer is parse-back-tested **100%** against the compiled lines on every export
- Built lawful discovery over **29** keyless ATS boards plus aggregator APIs across **18** country markets, with a work-authorization verdict at ingest — never scraping, never auto-submitting
- Enforced **776** automated gates including a **5,000-case** generated-input matrix, an encrypted cross-device vault, and a read-only demo that cannot spend a token
- Built a free-tier **LLM** router (Gemini and **Groq**, JSON-schema strict) with a deterministic fallback behind every feature, and a strategist pass that reads a whole posting into what the company values before it plans the page

Currently Building: Transformers Internals

ACHIEVEMENTS

- **1st Place** — **Agentic & GenAI Showcase**, Techgyan Hackathon, IIT Ropar — Won first place in the **Agentic & GenAI** showcase at IIT Ropar's Techgyan Hackathon with Sifarish, the evidence-compiled job-hunt system
- National NTSE Scholar — National Talent Search Examination scholar, Government of India (**2021**) — a national-level test of aptitude and reasoning

TECHNICAL SKILLS

AI & ML: Tesseract.js, Gemini, Evals

Languages: Python, TypeScript

Frameworks & Tools: FastAPI, Vite

PUBLICATIONS

- PRANA — A Layered Framework for Sustainable AI Compute (position paper, **v1**) — Reframes AI's energy-water crisis as exergy misplacement and proposes a five-layer stack, follow-the-cold routing across hemispheres, and the compute-boiler facility archetype for district heating